Activity 1 - DNA Extraction

DNA belongs to the group of acids originally found within the nucleus of the cells, hence the name nucleic acids.

We will extract DNA from fruit to investigate how it looks and feels. This procedure is similar to what scientists have to do before they can use the information contained in this DNA. This information can be used to improve crops so that they are more resistant to disease, insect invasion or changes in climate.



The various chemicals used allow us to separate DNA from the cell. First of all, the cell wall needs to break open. The soap/detergent/extraction solution ruptures the outer part of the cell, while the salt helps separate DNA from other cellular chemicals, such as carbohydrates. The process of cooling helps protect the DNA from enzymes that normally do not bother the DNA because the DNA remains separated from the rest of the cell by the nuclear membrane. The cold temperature slows down these enzymes. Filtering removes most solid matter in the mixture. The last cooling helps the DNA solidify and precipitate.

Objectives

- Extract DNA from plant cells
- Understand the general structure of cells

Materials

- 1-small ziplock bag
- 20 ml of extraction buffer
- Skinned kiwi fruit (each fruit cut into 8 pieces) or one large strawberry (each provides ~30 g per student pair)
- 500 ml beaker (class)
- Cheese cloth
- ice water bath
- Ice cold 95% ethanol (2 ml)
- 1 small test tube
- 1 glass hook
- Transfer pipettes



Procedures

1. Cut half of one kiwi fruit or the strawberry into pieces and place in the ziplock bag.
2. Add 20mL extraction solution in the zipper bag. Close bag and CAREFULLY squeeze out air.
3. Crush the kiwi/strawberry and extraction solution thoroughly for 5 minutes. CAREFUL don't break the bag!
4. Using the ice water bath, alternate cooling the mixture for one minute and continue to press the contents together for one minute. Repeat this three times
5. Use the cheese cloth or gauze in a funnel to filter the mixture.
6. Using the large transfer pipettes, aliquot approximately 2 ml of the kiwi/strawberry fruit solution into a test tube.
7. Add approximately 2 ml of ice-cold ethanol to each tube by dropping it slowly down the side of the test tube, allowing it to rest on top of the kiwi/strawberry fruit mixture. Do not agitate the solution.

8. Let the solution sit for two minutes without disturbing it. The DNA stands should form at the junction between the filtrate and the isopropanol and will appear as transparent, slimy, white mucus which can be carefully removed with the glass hook.

Technician preparation for experiment

Time Required: ~ 20 minutes

Night before put 95% ethanol in freezer

1. Make extraction solution (see below).
2. Start water heating to 60°C.
3. Prepare ice-water bath.
4. Prepare fruit pieces.
5. Gather materials for each student group as listed below.

Extraction Solution

Materials (100 ml)

- 10 ml of clear shampoo (Suave daily clarifying shampoo)
- 1.5 g of table salt
- Distilled H_2O

Procedure (modify amount depending on the size of a class)

1. Mix 90 ml of distilled water and 1.5 g of salt.
2. Add shampoo until solution volume is 100 ml. Stir slowly to avoid foaming of the shampoo.
3. Measure 20 ml of solution into 1L zipper bags (1 per student pair).

Procedure Questions

1. Why do we “crush” the kiwi/strawberry fruit?
2. Why do we use shampoo?
3. What does the salt do?
4. Why do we need to cool the mixture?
5. What does the cold ethanol do?
6. Why can’t we use room temperature ethanol?

Discussion Questions

1. To extract DNA from cells, what must you isolate it from in the case of a plant such as strawberry?

2. What steps did we use to extract the DNA?

3. What is DNA used for when it is extracted?

Answers to Procedure Questions

1. **Why do we “crush” the kiwi/strawberry fruit?** Crushing the kiwi/strawberry fruit physically breaks apart the cell walls.

2. **Why do we use shampoo?** After the cell walls have been disrupted during mechanical mashing of the fruit, the detergent in the shampoo disrupts the cell and nuclear membranes of each cell to release the DNA. It does this by dissolving lipids and proteins that hold the membranes together.

3. **What does the salt do?** The salt neutralizes the negative charges on the DNA and thus enables the DNA strands to stick together. It also causes proteins and carbohydrates to precipitate.

4. **Why do we need to cool the mixture?** DNases or restriction enzymes that destroy DNA are present in the cell’s cytoplasm. They are there to protect the cell from invasion by viruses. Once the nuclear membrane is destroyed by the soap, the DNA is now susceptible to the DNases and will quickly be degraded. However, these enzymes are temperature sensitive and cooling the solution slows down the process of degradation.

5. **What does the cold ethanol do?** Everything except the DNA will dissolve in ethanol. The ethanol pulls water from the DNA molecule so that it then collapses in on itself and precipitates. The DNA will become visible as white mucous strands that can be spooled with the wooden applicator stick.

6. **Why can’t we use room temperature ethanol?** The colder the ethanol is the greater the amount of DNA that is precipitated. (You could try having some of the students use room temperature ethanol and see if the amount of DNA they can spool is the same or less than that for the groups using the ice-cold ethanol.)

Discussion Questions and Answers

1. **To extract DNA from cells, what must you isolate it from in the case of a plant such as strawberry?** All the other parts of the cell - the cell wall, cell membrane, nuclear membrane, mitochondria, vacuoles, endoplasmic reticulum, Golgi apparatus, lysosomes, etc.

2. **What steps did we use to extract the DNA?** First we broke apart the cell walls by physically squishing the fruit. The chemical (detergent) process broke down the cell walls, cell membranes and nuclear membranes. The fruit mixture was cooled to stop the DNases released from the cytoplasm from destroying the cell's DNA. The mixture was filtered to separate out the large cell parts that are not needed. The DNA was then precipitated through chemical means (the ethanol).

3. **What is DNA used for when it is extracted?** DNA can be used for the identification of people involved in crimes, to help determine parentage of people and also of plants and animals, and to check for genetic defects. For example, the DNA of these kiwi/strawberry fruits can be compared to other samples to determine if one of them has been altered in some fashion, such as changes that might be made to make a crop more nutritious. DNA from one organism carrying a gene that codes for a specific trait might also be used for transformation. The section of DNA containing this particular gene can be inserted into a different organism so that the altered organism now has a specific trait that it did not previously carry.

(Adapted from http://www.ctbiobus.org)

References

www.accessexcellence.org
www.biotech.iastate.edu
www.biology.arizona.edu

The following website provides a protocol for extracting your own DNA!
http://www.nature.ca/genome/05/051/pdfs/DNAextract_e.pdf